

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Specification for Line Pipes
Document Number	EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-001
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Specification for Line Pipes

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Revision History

Rev No.	Date	Description of Revision
R01	12-07-2022	Issued For Review

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-001

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

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Abbreviations

Abbreviations	Meaning
ASME	American Society of Mechanical Engineers
API	American Petroleum Institute
CS	Carbon Steel
DEP	SPDC- Design and Engineering Practices
DP	Design Pressure
DWT	Drop Weight tear
FEED	Front End Engineering
HAZ	Heat affected zone
ISO	International Organisation for Standardization
LTO	License To Operate
3LPE	Three-Layer Polyethylene
MMscf/d	Million standard cubic feet per day
NDE	Non-Destructive Examination
NPS	Nominal Pipe Size
OD	Outer Diameter
PE	Polyethylene
PSL	Product Specification Level
TD	Design Temperature
SPDC	Shell Petroleum Development COMPANY Nig. Limited
SMTS	Specified Minimum Tensile Strength
SMYS	Specified Minimum Yield Strength
WT	Wall thickness
WPS	Welding Procedure Specification
WPQR	Welding Procedure Qualification Record

Site and Environmental Conditions

Refer to Basis of Design Document EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002

Definition of Terms

Terms	Definitions
BFPCL	Brass Fertilizer & Petrochemical Limited.
COMPANY	Party (Usually BFPCL) that initiates the project and ultimately pays for its design and construction. The COMPANY will generally specify the technical requirements. The COMPANY may also include an agent or consultant authorized to act for and on behalf of, the COMPANY.
Engineer	The party that carries out the FEED Engineering of the project.
Manufacturer/Supplier	Party, which manufactures or supplies equipment and services to perform the duties specified by the MANUFACTURER.
Shall	The word 'shall' indicate a mandatory requirement.
Should	The word 'should' indicate a recommendation.
Project	Front End Engineering Design for Gas Gathering Pipelines (Phase1)

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

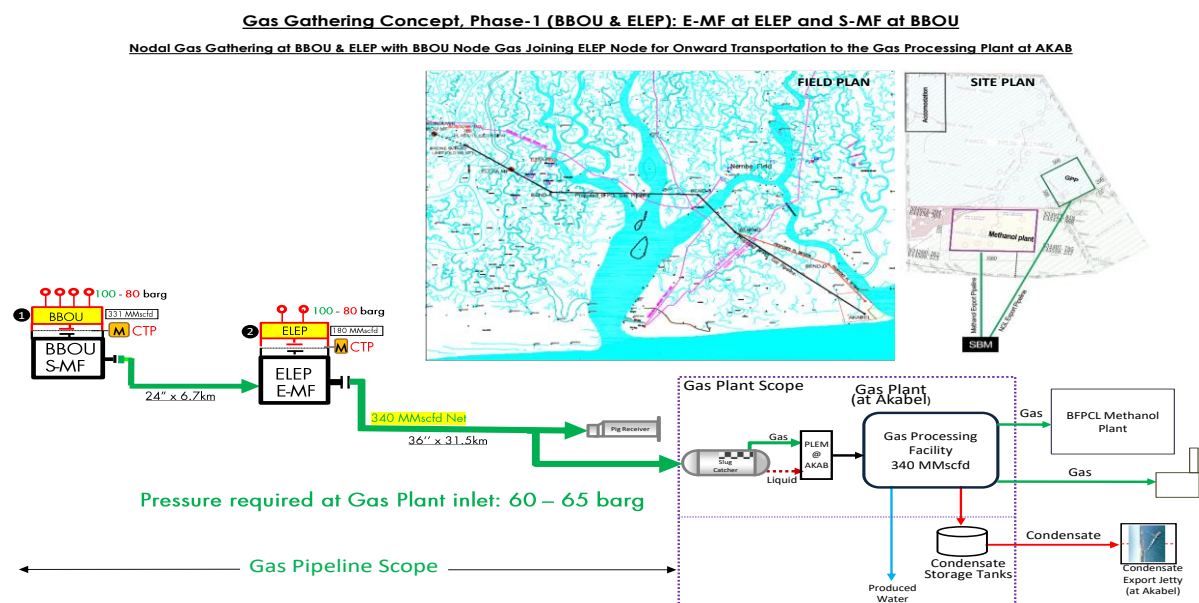


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This specification details the minimum requirements for manufacturing, inspection, testing, marking and delivery of and supply of Mainline carbon steel line pipes for phase 1 of BFPCL Gas Gathering Pipelines & Associated Infrastructure. The line pipe shall be produced in accordance with the requirements of the latest edition of API Spec 5L PSL 2 for a multiphase service.

The Pipes shall meet the supplemental requirements of this specification. Unless otherwise specified by the COMPANY, line pipes shall be of Grade X70 SAW - 100% radiographed.

This specification shall be read in conjunction with DEP 31.40.20.37-Gen.: Line pipe for Critical Service. Paragraphs or parts of paragraphs contained in ISO 3183: 2007(API 5L) which remain unaffected by the requirements of this specification, shall remain fully applicable, in so far these relates to Line pipes. In addition to the requirements of this specification, quantities and dimensions shall be as defined in the material requisitions list.

2. Regulations, Codes And Standards

The latest versions of the regulations, codes, and standards and extant amendment or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- COMPANY's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the COMPANY for resolution.

2.1. Project Technical Documents

Key Project Technical Specifications are listed below.

REF	DOCUMENT NUMBER	TITLE
[1]	EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
[2]	EA-BFPCL-GPMC-GGPL-FEED-PPL-RPT-002	Wall Thickness Calculation Report
[3]	EA-BFPCL-GPMC-GGPL-FEED-PPL-ALS-001 to 036	Pipeline Alignment Sheet
[4]	EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-009	Pipeline Construction and HDD Installation Specification.
[5]	EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-010	Pipeline Cleaning, Gauging, & Hydrotesting Specification.
[6]	EA-BFPCL-GPMC-GGPL-FEED-PPL-PHY-001	Pipeline Integrity Management Philosophy.

2.2. Shell Specifications

The following DEPs are applicable:

REF	DOCUMENT NUMBER	TITLE
[7]	DEP. 31.40.20.37-Gen	Line pipe for Critical Service (amendments/supplements to IOGP S-616 AND DNVGL-ST-F101 2019)
[8]	DEP. 82.00.10.10-Gen	Project Quality Assurance
[9]	DEP. 31.40.40.38-Gen	Hydrostatic Pressure Testing of new Pipeline (February 2017)
[10]	DEP. 61.40.20.30-Gen	Welding of Pipelines & Related Facilities (Amendments / supplements to API Std 1104)

2.3. International Codes and Standards

REF	DOCUMENT NUMBER	TITLE
[11]	ASME B31.8	Gas Transmission and Distribution Piping Systems
[12]	ASME B36.10M	Welded and Seamless Wrought Steel Pipe
[13]	API 5L	Specification for Line Pipe
[14]	ASME B36.10M	Welded and Seamless Wrought Steel Pipe
[15]	API RP 5LW	Recommended Practice for Transportation of Line pipe on Barges and Marine Vessels
[16]	API RP 5L1	Recommended Practice for Railroad Transportation of Line Pipe
[17]	BS EN 10204	Metallic Products - Types of Inspection Documents
[18]	ISO 3183: 2007	Petroleum and Natural Gas Industries Steel Pipe for pipeline transportation systems.
[19]	ISO 9001	Quality Systems Model for Quality Assurance in Design, Development, Production, Installation and Servicing

3. Line Pipe Data

3.1. 24" Multiphase Pipeline Data

The 24" pipeline design data for the Multiphase Pipeline Project are as shown below

Parameter	Description
Design Code	ASME B31.8
From	BBOU Manifold Station
To	ELEPA Manifold Station
Pipeline Length	6.7km
Design Life	30 years
Process Fluid	Condensate / Saturated Hydrocarbon Gas.
Design Pressure	110 barg
Operating Pressure	100 barg
Max Design Temperature	60°C
Min Design Temp (Pipeline)	12.7°C
Min Design Temperature (Vent Line)	-13.3°C
Line Pipe Material Grade	API 5L-X70(PLS2) SAWL - 100% radiographed.
Installation Temperature	23°C
Specified Minimum Yield Strength (SMYS)	483 Mpa
NPS	24"
Density	7,850 kg/m ³
Young's Modulus of elasticity	2.07 x 10 ⁵ MPa
Poisson's ratio	0.3
Coefficient of linear thermal expansion	11.7 x 10 ⁻⁶ m/ m/ °C
Ultimate Tensile Strength	570 MPa
Steel Specific Resistivity	0.018 Ω m/mm ²
Wall Thickness	15.88mm
Design Factor	0.6
Corrosion Allowance	3 mm
ANSI Class	900#
Parent Pipe External Coating	3LPE
Concrete Coating Thickness	60mm
Installation Condition	Trenching
Cathodic Protection	Impressed current

3.2. 36" Multiphase Pipeline Design Data

The 36" pipeline design data for the Multiphase Pipeline Project are as shown below

PARAMETER	DESCRIPTION
From	ELEPA Manifold Station
To	AKABEL Station
Pipeline Length	31.5km
Design Life	30 years
Process Fluid	Condensate/hydrated Hydrocarbon Gas.
Design Pressure	110 barg
Operating Pressure	100 barg
Max Design Temperature	60°C
Min Design Temp (Pipeline)	12.7°C
Min Design Temperature	-13.3°C
Line Pipe Material Grade	API 5L-X70 (PLS2) SAWL - 100% radiographed.
Installation Temperature	23°C
Specified Minimum Yield Strength	483 Mpa
NPS	36"
Density	7,850 kg/m ³
Young's Modulus of elasticity	2.07 x 10 ⁵ MPa
Poisson's ratio	0.3
Coefficient of linear thermal expansion	11.7 x 10 ⁻⁶ m/ m/ °C
Ultimate Tensile Strength	570MPa
Steel Specific Resistivity	0.018 Ω m/mm ²
Wall Thickness	23.83mm
Design Factor	0.6
Corrosion Allowance	6 mm
ANSI Class	900#
Parent Pipe External Coating	3LPE
Concrete Coating Thickness	85mm
Installation Condition	Trenching
Cathodic Protection	Impressed current

4. Material Requirements

4.1. General

Line pipes for the gas export pipeline shall be of PSL 2 grade. They should be of the correct diameter. The material should be supplied in the quenched / tempered condition and as specified in the line pipe data sheets provided with the Purchase Order.

The steel shall be made by either the basic oxygen or electric arc process and shall be fully killed and fine grain treated.

Several finished straight pipes specified by the COMPANY shall be made suitable for fabrication into bends by hot forming / induction bending and, in such cases, shall possess the necessary dimensional and physical properties to satisfy these requirements considering wall thinning and material strength reduction or any other changes that may occur during the bending process.

The pipe material shall be suitable for manual, semi-automatic or automatic welding using conventional electrodes and filler wires. The welded joints shall be capable of meeting the mechanical properties of the parent pipe material as a minimum.

4.2. Qualification Trials

A detailed line-pipe Manufacturing Procedure Specification (MPS) shall be submitted by the SUPPLIER for COMPANY review / approval, in accordance with sections B.1, B.3 and B.5 of Annex B of ISO 3183 as amended by sections B.1, B.3 and B.5 annex B of DEP 31.40.20.37.

The SUPPLIER shall provide evidence of the weldability of the line pipe according to DEP 61.40.20.30- Gen requirements of pipe produced by the proposed manufacturing route to the proposed size and target chemistry as indicated in the Line-pipe Data Sheets.

4.3. Chemical Properties

The chemical composition shall conform to table 1, below in accordance with the requirements of ISO 3183 (Table J1 Annex J).

Table 1: Required Chemical Composition for Line Pipe

Steel grade	Mass fraction, based upon heat and product analyses (maximum) %.									Carbon equivalent a (maximum) %	
	C	Si	Mn	P	S	V	Nb	Ti	Other	CE _{IW}	CE _{PCM}
X70QO	0.17	0.45	1.75	0.020	0.010	0.10	0.05	0.06	e, h	0.42	0.23

Notes:

e= the sum of niobium, vanadium and titanium concentration shall be $\leq 0.06\%$.

h= Cu $\leq 0.50\%$, Ni $\leq 0.50\%$, Cr $\leq 0.50\%$, Mo $\leq 0.50\%$; B $\leq 0.0005\%$.

4.4. Chemical Composition Criteria

Based upon product analysis, the CE_{IW} limits apply if C > 0,12 % and the CE_{PCM} limits apply if C $\leq 0,12\%$.

Note:

CE_{IW} = Carbon equivalent based on international institute welding equation

CE_{PCM} = Carbon equivalent based upon the chemical portion of the Ito Bassyo carbon equivalent.

For each reduction of 0,01 % below the specified maximum for C, an increase of 0,05 % above the specified maximum for Mn is permissible, up to a maximum increase of 0,20 %

Al_{total} $\leq 0,060\%$; N $\leq 0,012\%$; Al/N $\geq 2:1$ (not applicable to titanium-killed steel or titanium-treated steel).

- Nb + V + Ti $\leq 0,15\%$.

Since it is PSL2 pipe and with a product analysis maximum carbon mass fraction of 0.16 %, the chemical equivalent, CE_{IW}, shall be determined using equation below:

$$CE_{IIW} = C + \frac{Mn}{6} + \frac{Cr + Mo + V}{5} + \frac{Ni + Cu}{15}$$

The symbols for the chemical elements in above equation. represent the mass fraction in percent stated in table 2, above.

For each order, the Supplier shall propose a chemical composition for the pipe to be supplied.

Product chemical analysis shall be as specified in table 1 of this specification which is in accordance with table J1 of Annex J of ISO 3183.

The Manufacturer/Supplier's target product analysis shall be shown in the Manufacturing Procedure Specification. A single value target of CE_{IIW} or CE_{pcm} shall be stated. The range of elements shall be such that the variation in the single value target CE_{IIW} does not exceed ± 0.03 units or CE_{pcm} does not exceed ± 0.02 units with the following additional restrictions in table 2 below.

Table 2: Table of additional Restrictions

C	Mn	Si	Ni	Cu	V	Nb	Al	Ti	Mo	Cr
0.03 (HFW and SAW)	0.3	0.25	0.1	0.1	0.03	0.02	0.04	0.015	0.05	0.05

NOTE: The above figures represent the total range and not a plus or minus tolerance.

No changes in the chemical composition (within the tolerances listed above) are permitted from the approved MPS composition. No intentionally added elements shall be increased (beyond the limits above) or removed.

The chemical composition recorded for the pipes used in first-day production testing shall set the reference CE. Where any of the above elements is not intentionally added, the manufacturer may propose maximum values for approval by the OWNER. In this case the above ranges shall not apply.

The chemical analysis and reporting in inspection documents shall include all elements covered by the requirements of table 2 above in accordance with Annex J Table J1 of ISO 3183.

4.5. Mechanical Properties.

The line pipe shall conform to the mechanical properties determined by the relevant tests as outlined in section 9 of ISO 3183 as amended by DEP 31.40.20.37. Acceptance criteria for each mechanical property shall be as outlined but not limited to this specification in accordance with ISO 3183 as amended by DEP 31.40.20.37.

4.5.1. Tensile Strength and Yield Strength

Pipe Grade	Yield Strength Rt0.5 (MPa)		Tensile Strength Rm		Ratio	Pipe weld seam Tensile
	Minimum	Maximum	Minimum	Maximum	Maximum	Minimum
X70Q	485	635	570	760	0.93	570

4.5.2. Hardness

The hardness in the pipe body, weld and HAZ shall not be greater than 270 HV10 for the X65 grade. Hardness tests shall be carried out in accordance with requirements specified in section J.8.3.2 annex J of ISO 3183 as amended by DEP 31.40.20.37.

4.5.3. Charpy Impact Resistance

The required Charpy Impact Resistance requirements for the pipe body to avoid brittle fracture shall be:

- Required Energy: 45 J
- Test temperature: (Minimum design temperature as specified on pipeline PEFS and process engineering blowdown report)

Test locations, frequency, and test requirements for the weld and HAZ shall be as detailed in DEP 31.40.20.37 and ISO 3183.

Transverse Charpy impact tests shall be carried out for the avoidance of long running shear fracture as stated in section 9.8. and 10.2.3.3 of ISO 3183 and modified by DEP 31.40.20.37 Required Energy: will be as stated in DEP 31.40.20.37 (for full size, transverse specimen).

- Test temperature: (Minimum design temperature as specified on pipeline PEFS and process engineering blowdown report).

4.5.4. Drop Weight Tear (Fracture Toughness)

The required Drop Weight Tear tests shall be carried out in accordance with ISO 3183 as modified by DEP 31.40.20.37. The test temperature shall be at the minimum design temperature for the pipeline as stated on the PEFS. The average shear fracture shall not be less than 85% with no individual specimen less than 50% shear area.

SUPPLIER shall conduct additional qualification trials at the applicable temperatures.

4.5.5. Flattening

The line pipe shall be able to be flattened to 50% of the original OD without cracks or breaks in either weld or parent metal. Test specimens from the line pipe shall be used to assure this and shall be further flattened to 1/3 of original OD without cracks or breaks other than in the weld.

4.5.6. Guided-bend Test

The line pipe test pieces shall not fracture completely, reveal any cracks or ruptures in the weld metal longer than 3,2 mm (0.125 in), regardless of depth, or reveal any cracks or ruptures in the parent metal, HAZ or fusion line longer than 3,2 mm (0.125 in) or deeper than 12,5 % of the specified wall thickness

5. Manufacturing Requirements

5.1. Manufacturing Process

The manufacturing process shall be in accordance with DEP 31.40.20.37 table 3 and ISO 3183. It is recommended that the line pipes are manufactured by a HFW (high frequency electric welding) or Longitudinal SAW (sub-merged arc welding) process since a seamless process is not feasible because of the pipeline size.

The pipes shall be manufactured strictly in accordance to the above standard and DEP and the following requirements:

- Vendor shall submit appropriate WPS/MPS and MPQR/WPQR for review / approval.
- Some quantities of the pipe manufactured to the requirements of this specification shall be used as mother pipe for induction bends.
- Any change in the essential variables during the manufacturing process shall require a re-qualification of the manufacturing procedure specification (MPS)
- Additional essential variables apply when HFW is the pipe manufacturing process and these should also be complied with.

5.2. Dimensional Requirements and Tolerances

Dimensions, weights and lengths shall be in accordance with ISO 3183, section 9.11 and tested in accordance with section 10.2.8.

Pipes shall be supplied in accordance with the length requirements in DEP 31.40.20.37-Gen.

5.3. Machining

All carbon steel pipes whose ends are to be bevelled shall be performed either by machine or flame –cutting and the process will ensure that the ends are free of burr. In case of flame-cutting, the ends shall be grinded or machined to remove any trace of melted metals.

The inside reinforcement of the weld must be ground-off at each end of the pipe. This machining shall be done over a length of 10 cm and must leave a gradual transition in accordance with (API 5L).

Pipes shall be supplied in lengths of 12.20 m $\pm$ 0.60 m for sizes 6" and above. The average length of each item of the order shall be 12.00 m minimum. Unless otherwise specified in the Purchase, some of the pipes may have an individual length between 9.00 m and 11.59 m,

For sizes less than 4", the average length shall not be less than 5.33m (17.5 ft).

The square pipe ends shall be cut perpendicular to the pipe longitudinal axis within 1.6 mm across any diameter.

All pipe lengths shall be recorded.

Joining by welding of two pipe sections to produce the specified pipe length is not permitted.

The deviation from a straight line (taut wire) used in straightness checking of the line pipes 0.15% over the total length of the pipe.

The taut wire shall be positioned at 3 or 9 O'clock along the centreline axis of the line Pipe.

When necessary, cold straightening is allowed provided that the resulting permanent straining does not exceed 1.5%.

5.4. Pipe End Preparation

All pipes shall be furnished with both ends bevelled in accordance to API 5L for welding to an angle of 30° with a tolerance of 0° to +5° measured from a line drawn perpendicular to the axis of the pipe. The bevel root face width shall be 1.6 mm (0.063 in), with a tolerance of ± 0.8 mm (0.031 in).

Bevelling shall be carried out by straight machining only. Internal taper shall be in accordance to ISO 3183 table 13 (when measured in the pipe longitudinal axis, from base of root face to inside surface of pipe) and that radial height of taper does not exceed 2 mm, at any location around pipe end.

When internal taper is intended to be machined, VENDOR shall submit to COMPANY, a procedure for the checking of the remaining wall thickness at the pipe ends after machining.

5.5. Surface Condition

Pipes shall be free from unacceptable indications and protrusions and shall have good quality finish indicative of good workmanship. Minor indications may be removed by grinding provided that a smooth curved surface is maintained and that the wall thickness is not decreased below the specified thickness.

Pipe surface must be substantially free from loose mill scales, scabs, pits, scores and laps as well as un-repaired tears, raised slivers, bristles of steel, sharp notches, scratches or mechanical marks.

6. Quality Control

Supplier is required to submit with quotation his proposed inspection data sheet, which shall cover all requirements indicated in the Inspection Data Sheet and any other applicable referenced documents. Materials, design, manufacturing, and preparation for shipment of the pipes shall be subject to inspection by contractor and/or COMPANY or their appointed third-party inspection agency designated in the contract and, where applicable, by certification authority in the supplier's manufacturing plant.

The supplier shall assume overall responsibility for the design, engineering, fabrication, and testing procedures of the pipes and shall provide a complete guarantee to cover these activities. The Supplier shall maintain quality records of all tests, inspection and measurements detailed in this document and the Contractor approved quality plan as documentary evidence of conformance to quality requirements. Quality records shall be made available to Contractor's representative for analysis and review.

7. Quality Assurance And Quality Control

SUPPLIER shall show that an effective quality assurance system is in operation for both product and services and this shall comply with ISO 9001 or equivalent as approved by COMPANY.

The SUPPLIER shall establish and maintain a fully documented Quality Assurance /Quality Control system in accordance with ISO 9001 to ensure:

- Effective inspection and objective evidence that items conform to Purchase Order requirements
- Adequate identification and suitable handling of items.

The SUPPLIER shall submit on request, documentary evidence of any Quality Reviews to ISO 9001, and internal system audits and corrective action requests.

All certification, test results, reports and any other documentation submitted to the COMPANY relative to the inspection system and results of inspections shall be in the English language.

Any verification of quality performed by COMPANY, or its representative will not relieve the SUPPLIER in any way of its responsibility to produce acceptable items nor shall it preclude any subsequent rejection of items.

COMPANY shall establish mandatory hold points where applicable, which require verification of selected characteristics of any items or process by COMPANY and beyond which the manufacture shall not proceed until released.

The SUPPLIER shall establish and maintain a system of traceability, which complies with ISO 9001. The system shall be such that each line-pipe is allocated a unique number, and this number shall be maintained such that at any point the individual line-pipe can be traced back to its original heat number. These numbers will be in blocks allocated by the COMPANY to the SUPPLIER, and the SUPPLIER shall demonstrate its ability to maintain traceability of such numbers throughout production.

8. Inspection and Testing

8.1. General

Inspection and Testing of pipe after manufacturing shall be done in accordance with tables 18 and 20 sections 10.2.1, 10.2.2 and 10.2.3 of ISO 3183 as amended by DEP 31.40.20.37-Gen. Non-destructive testing shall be carried out in accordance with ISO 3183 annex E. All procedures proposed for non-destructive testing shall be submitted for COMPANY's approval.

Prior to the commencement of manufacture, the SUPPLIER shall prepare a written Inspection and Test Plan, which describes the inspection to be performed on each product. The Inspection and Test Plan, reference procedures and changes thereto are subject to the approval of the COMPANY. SUPPLIER shall provide the following in addition to the Inspection and Test Plan:

- A flowchart illustrating each manufacturing point affecting the quality of the product and its relative location in the production cycle where conformance of characteristics shall be verified. The SUPPLIER shall include all inspection points of its own verification of quality.
- The characteristics to be inspected at each inspection point from procedural approval to shipping inspection. The procedures shall be submitted to COMPANY for review and approval prior to production commencement.

8.2. Quality Records

The SUPPLIER shall maintain quality records of all tests, inspection and measurements detailed in this document and the COMPANY approved quality plan as documentary evidence of conformance to quality requirements. Quality records shall be made available to COMPANY's representative for analysis and review.

Quality records shall include item identification by reference to revision number, acceptance criteria, specific inspection performed, and results obtained (If measurements are not a requirement, include in the record the basis of acceptance), date of inspection, identification of inspector, data recorder charts, qualification of material, personnel, procedures, and equipment.

The SUPPLIER shall perform trend analyses of all factors affecting the quality of the product.

8.3. Hydrostatic Tests

Pipe shall be sized, straightened and hydrostatically tested in accordance with API 5L or section 10.2.6 of ISO 3183.

SUPPLIER shall maintain a record of hydrostatic test for each pipe and shall certify that every length of delivered pipe has been tested in accordance with specified requirements. The test pressure shall be held for not less than 20 seconds.

The test information shall be recorded on a suitable recording chart. The serial number of the pipe being tested shall be legibly recorded at the side of the pressure trace on the chart. In cases where a pressure test is discontinued because of a temporary failure of the test equipment, the serial number of the pipe under test at the time of failure shall be recorded on the chart and it shall be shown clearly that the pipe has been retested and has satisfactory results.

The test pressure shall be as specified in Section 10.2.6.5 of IOGP S-616, with a minimum of 95 % of minimum yield strength based on pipe thickness at the lowest manufacturing tolerance.

During hydrostatic testing, each length of pipe shall withstand, without leakage a hydrostatic test pressure calculated on the basis of minimum specified wall thickness in accordance to API 5L.

$$t = PD/2SFET$$

Where;

P. Hydrostatic strength test pressure (barg)

D. Outside diameter of pipe (mm)

S. Specified Minimum Yield Strength (SMYS) (N/mm²)

t. Specified Wall Thickness (mm)

F. Design Factor

E. Longitudinal Joint Factor

T. Temperature Derating Factor

If end sealing rams that produce additional compressive longitudinal stresses apply, the end load compensation factor as determined by the formula given in API 5L, Clause 10.2.6.6 shall be used.

8.4. Final Inspection

The SUPPLIER shall inspect the final item to ensure compliance with contract requirements. A check shall be made of all inspection records to verify that items were inspected at all points shown in the Inspection and Test Plan and that these records are complete and available to COMPANY's representative. Visual inspection and dimensional checks shall be performed both at the point of

discharge and receipt to confirm that no damage took place during transportation.

8.5. Verification

All SUPPLIER inspection systems may be subject to evaluation and surveillance by COMPANY's representative to ensure that they meet the requirements of this standard and Purchase Order. The SUPPLIER's operations required by this standard are subject to:

- Procedure compliance checking at unscheduled intervals to determine that the SUPPLIER's inspection system is being effectively applied.
- Product verification to determine compliance with contract requirements. The method of verification is at the discretion of COMPANY's representative.

All certifications and qualifications shall be certified by an independent certifying authority, such as Lloyds, DNV, etc.

8.6. Accommodation and Assistance

The SUPPLIER shall provide COMPANY's representative with office accommodation and facilities required for the proper accomplishment of his work and shall provide any reasonable assistance required by COMPANY's representative for verification, documentation, or movement of items.

COMPANY's representative shall be afforded unrestricted opportunity to perform verification of inspection. The SUPPLIER shall make available its inspection equipment for use by the COMPANY's representative to verify conformance to contract requirements.

The SUPPLIER personnel shall be made available for the operation of inspection equipment as reasonably required by COMPANY.

8.7. Pipe Coating

Pipe shall not be coated unless specified in the purchase order. In the case coating is required before delivery to COMPANY, the pipe SUPPLIER shall prepare and submit detailed coating procedures to COMPANY for approval prior to the commencement of the order. The procedures shall be according to the provisions of the project anti-corrosion coating material specification.

8.8. Defects

8.8.1. Repair of Defects

The COMPANY shall be advised of each and every line pipe that is designated for rectification work. Such line pipe spools shall be marked as defective and taken to a special quarantine area. Pipe shall not be removed from this area without the permission of the COMPANY in writing. SUPPLIER shall issue a repair procedure to be approved by COMPANY before production works commences.

8.8.2. Rejected Line Pipes

The COMPANY shall be advised of all line pipes that have been rejected and taken away from the original stockpiles. Such line pipe spools shall be clearly marked as rejected and moved into a special quarantine area. Pipes shall not be removed from this area without the expressed permission of the COMPANY in writing.

9. Acceptance

The pipes supplied will not be accepted by the COMPANY unless the material certificates have been submitted in advance. In addition to checking the certificates, the COMPANY shall check the quantities presented and shall have checked, by sampling, the following characteristics:

- Any radiographic and ultra-sonic examinations reports
- Shapes and nominal dimensions
- Dimensional tolerances
- Surface treatment and finish conditions
- Marking and finishing conditions.

10. Preparation For Shipment

10.1. Marking

Marking shall be in accordance with this specification and ISO 3183, section 11. In addition, the words "BFPCL GAS GATHERING PIPELINES & ASSOCIATED INFRASTRUCTURE PROJECT" shall be included in the stencil area. All markings shall be performed at the pipe mill unless otherwise agreed to in writing by COMPANY. Each pipe shall be uniquely identified.

Unless specified otherwise on the purchase order paint marking shall be located as follows:

- Inside Surface (Not less than 150mm from one pipe end).

Pipe markings shall include the following information:

- Specification and grade of material
- name of the manufacturer
- International Standard designation number
- Specified outside diameter of Pipe
- Specified wall thickness of Pipe
- Specified and grade of material
- Product specification level
- Type of Pipe
- Product ID Number
- Batch number.

However, any pipes less than 3 mm thick shall not be marked by stamping but by etching.

11. Transportation

11.1. General

The SUPPLIER shall submit to the COMPANY for review and approval, information detailing the proposed method of handling, stacking during storage, and stacking and securing Line pipe for shipment. The stacking information shall include the maximum permissible stacking height for the line pipes.

The SUPPLIER shall also detail storage requirements for short (6 months) and long-term storage including any special maintenance procedures that may be required.

No welding of temporary attachments for handling, stacking or securing shall be permitted. All handling, loading and unloading shall be performed in such a manner as to avoid mechanical damage and corrosion.

11.2. Bevel Protectors

Bevel protectors of a type approved by the COMPANY shall be supplied and installed by the SUPPLIER before stockpiling of the line pipes takes place.

11.3. Loading and Stacking

Rail cars, trucks, lighters, ships or other conveyances shall be cleaned of debris and any substance that might damage the line pipes prior to loading. Suitable timber cradles or other packing shall be used to protect the line pipes and coating if applicable against damage in transit. Loading on or into rail cars, trucks, lighters, ships or other conveyance shall be performed in accordance with API RP 5L1 and API RP 5LW as appropriate. Pipe stacks shall not exceed 3 meters in height. In all cases loading shall be in accordance with a procedure approved by the COMPANY.

In addition to securing the pipes on the truck to prevent sideways movement, provision shall be made to prevent longitudinal movement to resist forces due to the vehicle braking, accelerating, or negotiating an incline.

Loading and unloading of line pipes shall be witnessed by COMPANY's representative whose decision on the suitability of protection / handling is final.

11.4. Protection

The SUPPLIER shall propose to the COMPANY for approval, suitable methods for the protection of internal surfaces against moisture during transportation. Line pipes shall not rest on projections, which could result in point stresses or be allowed to rub on an adjacent object. Loading and stacking shall be limited to prevent stresses, which result in out-of-roundness.

No pipe shall be shipped above deck, on sea going open barges or in such a manner that it is exposed to saltwater spray or atmosphere. If it is determined that the delivered pipe is chloride contaminated, the pipe SUPPLIER will reimburse COMPANY for the cost of removing the chloride contamination.

11.5. Packing

All the pipes shall be properly packed to avoid any damage during transportation and storage. Storage shall take place in a specially designated area, where the material will be stacked in a safe, orderly manner, to eliminate the risk of damage,

12. Documentation

All certification and documentation shall be as stated in ISO 3183 certification requirements except as amended by DEP 31.40.20.37.Gen.

Materials shall be delivered by the SUPPLIER with certificates from a classification society in accordance with BS EN 10204 3.2. The certificates and associated documentation supplied to COMPANY shall contain all relevant information, including, but not restricted to, the following:

- SUPPLIER's Identification
- COMPANY's Name and Order Number
- Metallographic Examination Records
- Description of Product, Dimension, Identification Number, Weight
- Steel Making Process
- Chemical Composition
- Heat Number
- Identification of Steel type and Grade
- Yield - Tensile Ratio
- Charpy Impact Results
- Hardness Tests
- Heat Treatment Condition and Records
- Hydrostatic Test
- NDT Procedures and Results
- Surface Inspection
- Dimensional Control Checks
- Manufacturing Procedure and Qualification Tests
- COMPANY's Line pipe Release Note.
- SUPPLIER's Third Party BS EN 10204 3.2 Certificates.

The SUPPLIER shall also supply a table of information stating the heat number, length, weight, testing, and inspection associated with each line pipe number.

13. Certification

Inspection Certification shall be in accordance with BS EN 10204 3.2 and shall be supplied as specified in SDRL. Certificate 3.2 shall be issued. Certificates shall be fully traceable to the source material.